🏋️ Exercise 1: Online Library System (Simple)

Scenario

A **University Online Library System** allows students to manage their book interactions. A **Student** can **Search for Books** by title or author, **Check Out a Book** (if available), and **Return a Book**. An **Librarian** is responsible for **Adding New Books** to the catalog and **Removing Books** from the catalog when they are lost or outdated.

Instructions

1. Identify the **Actors** (users/systems interacting with the system).
2. Identify the main **Use Cases** (the functions or activities performed).
3. Draw the **System Boundary** (the rectangle representing the system).
4. Connect the Actors to their respective Use Cases.

💻 Exercise 2: ATM System (Intermediate)

Scenario

The Automated Teller Machine (ATM) system serves Customers and a Bank Technician.
A Customer must first Insert Card and Authenticate (enter PIN) before performing any other action. A Customer can then Check Account Balance, Withdraw Cash, or Deposit Funds.
The Bank Technician is responsible for Refilling Cash in the machine.

Instructions

1. Identify the **Actors** (Customer, Bank Technician).
2. Identify the main **Use Cases** (Insert Card, Authenticate, Check Balance, etc.).
3. Draw the **System Boundary** (the ATM).
4. Use the **$<<include>>$** relationship to show which Use Cases *must* be performed as part of another (e.g., Authentication is always included when a customer wants to check their balance).
5. Connect the Actors to their respective Use Cases.

☕ Exercise 3: Coffee Shop Order System (Complex)

Scenario

The "Quick-Cup" Coffee Shop Order System manages customer orders. A Customer can Place an Order. The order process involves the Select Items step, and optionally, the customer can Apply Loyalty Discount.
The Barista receives the orders and Prepare Coffee. Once the coffee is ready, the Barista can Mark Order as Complete.
There is an Inventory System that automatically Updates Stock Levels every time an order is placed.

Instructions

1. Identify the **Actors** (Customer, Barista, Inventory System).
2. Identify the Use Cases (Place an Order, Prepare Coffee, Update Stock Levels, etc.).
3. Draw the **System Boundary**.
4. Use the **$<<include>>$** relationship where one Use Case is required for another (e.g., Selecting Items is included in Placing an Order).
5. Use the **$<<extend>>$** relationship for optional behavior (e.g., Applying Loyalty Discount extends the Place an Order Use Case).
6. Connect the Actors to their respective Use Cases.